

FIG. 7. The supply curve slopes upward from left to right.

WHAT MOVES THE SUPPLY CURVE

Determinants of supply.

A change in price moves us along the curve. Other causes shift the whole curve. The main ones are: the cost of inputs, such as flour and labour; technology, which lowers cost; taxes and subsidies; the prices of other goods the seller could make instead; the sellers’ expectations of future prices; and the number of sellers. Cheaper flour moves the curve to the right: sellers offer more at every price. A tax on bread moves it to the left.

EXCEPTIONS TO THE LAW

Exceptional supply curve a supply curve that does not slope upward in the usual way.

A few cases break the rule.

An **exceptional supply curve** does not rise in the usual way. There are several cases. The supply of labour may bend backward: at very high wages, a worker may choose more leisure and work fewer hours. Goods with fixed supply, such as old paintings or land in a city, cannot rise with price. Perishable goods, such as fish at the end of the day, may be sold at any price rather than wasted. If sellers expect the price to rise further, they may hold stock back and offer less now.

ELASTICITY OF SUPPLY

Price elasticity of supply the percentage change in quantity supplied divided by the percentage change in price.

How strongly sellers respond.

Price elasticity of supply means the percentage change in quantity supplied divided by the percentage change in price. In symbols, $E_s = (\% \text{change in } Q_s) / (\% \text{change in } P)$. The answer is positive, because price and quantity move together. If E_s is above 1, supply is **elastic**; below 1, **inelastic**; at 1, unitary. If E_s is zero, supply is **perfectly inelastic**; if infinite, **perfectly elastic**.

WORKED EXAMPLE. Elasticity of supply

GIVEN	From the supply schedule: at Tk 30 supply is 120; at Tk 35 supply is 150.
FIND	E_s for the rise from Tk 30 to Tk 35.
FORMULA	$E_s = (\% \text{change in } Q_s) / (\% \text{change in } P)$.
STEPS	Change in $Q_s = 150 - 120 = 30$. $\% \text{change} = 30/120 \times 100 = 25\%$. Change in $P = 5$. $\% \text{change} = 5/30 \times 100 = 16.7\%$. $E_s = 25 / 16.7 = 1.5$.
ANSWER	$E_s = 1.5$. Supply is elastic.
CHECK	Check: quantity rose by a larger percentage than price, so E_s is above 1. It is.

A neat fact: any straight supply line that starts at the point of origin has an elasticity of exactly 1 at every point. For such a line, quantity is a fixed multiple of price, so a 10 per cent rise in price gives a 10 per cent rise in quantity. A line that cuts the vertical axis (as our schedule does) has elasticity above 1. A line that cuts the horizontal axis has elasticity below 1.

A line from the origin.

Three factors matter. The first is spare capacity: a bakery with an idle oven can expand fast. The second is the ability to store output: goods that can be stored, such as rice, have elastic supply, and goods that spoil do not. The third is time: supply is inelastic in the very short run, when inputs are fixed, and more elastic in the long run, when firms can build and hire.

What decides the size?

CASE FILE: CAPACITY, 1 JUNE 2024

The bakery adds a second shift and can raise output from 120 to 150 loaves a day when the price rises from Tk 30 to Tk 35, as in the schedule. Its own elasticity of supply over that range is 1.5.

SUMMARY. Supply is the quantity sellers will offer at each price. The law says price and quantity rise together. Costs, technology, taxes, related prices, expectations and number of sellers shift the curve. Elasticity of supply measures the response; spare capacity, storability and time decide its size.

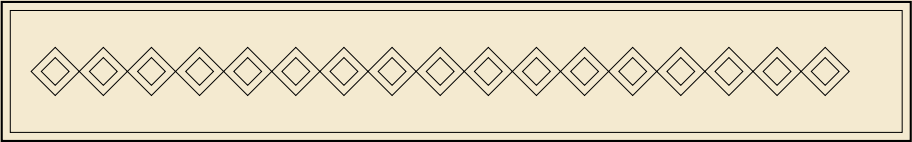
THINK IT THROUGH. Using $Q_s = 6P - 60$, find supply at Tk 25. Then say what happens to the curve if flour becomes cheaper.

WHERE PEOPLE GO WRONG. “A higher price shifts the supply curve to the right.” It does not; it moves us along the curve. Only a change in cost, technology and the like shifts it.

IN HISTORY. In 1890 Marshall compared supply and demand to the two blades of a pair of scissors. He said that it is as useless to ask which blade cuts as to ask whether price is set by supply or demand alone.

REVIEW QUESTIONS

1. State the law of supply.
2. Name two causes that shift supply to the right.
3. What is E_s of a straight supply line through the origin?



CHAPTER THE SEVENTH

MARKETS AND EQUILIBRIUM

How does a price find itself?

The chapter begins with a bakery that tries a price too high and a price too low. It defines a market, shows how supply and demand meet at an equilibrium price, describes the forces that bring it about, and asks whether lower prices are always good.

Suppose Rupali Bakery tries a loaf price of Tk 40 on Monday. At that price buyers want 60 loaves and sellers offer 180. On Tuesday it tries Tk 20. Buyers want 180 but sellers offer 60. Neither price lasts. On Wednesday the price settles at Tk 30, and buyers and sellers are both content. Why does Tk 30 work, and who set it?

Start with a question you can almost answer.

WHAT A MARKET IS

A **market** means any arrangement that brings buyers and sellers of a good together so that they can trade. It need not be a place. A phone call, a website or a bazaar can be a market. Markets differ by area (local, national, world), by time (very short, short and long period) and by the amount of competition. A **competitive market** has many buyers and many sellers, so that no one of them can set the price. Other kinds, monopoly and monopolistic competition, are studied in Chapter 10.

Market any arrangement that brings buyers and sellers of a good together to trade.
A market is a meeting, not necessarily a place.

EQUILIBRIUM

Equilibrium means a state in which nothing pushes the situation to change. In a market, the equilibrium price is the price at which quantity demanded equals quantity supplied. At that price, every buyer who wishes to buy finds a seller, and every seller finds a buyer.

Equilibrium a state in which quantity demanded equals quantity supplied, so there is no pressure for change.
Equilibrium is a point of rest.

WORKED EXAMPLE. Finding the equilibrium

GIVEN	Demand: $Q_d = 300 - 6P$. Supply: $Q_s = 6P - 60$.
FIND	The equilibrium price and quantity.
FORMUL A	Set $Q_d = Q_s$.
STEPS	$300 - 6P = 6P - 60$. Add $6P$ to both sides and add 60 : $360 = 12P$. So $P = 360 / 12 = 30$. Put $P = 30$ in demand: $Q = 300 - 180 = 120$.
ANSWER	Equilibrium price is Tk 30 and quantity is 120 loaves.
CHECK	Check in supply: $Q_s = 6 \times 30 - 60 = 120$. Both sides agree.

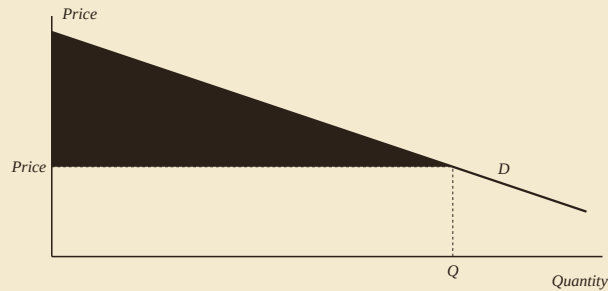


FIG. 5. Consumer's surplus is the shaded triangle between the demand curve and the price line.

Reading the figure.

The demand curve shows the most buyers would pay for each unit. The price line shows what they pay. The gap between the two, summed over all units bought, is the shaded area. If the price falls, the triangle grows, and buyers gain more. This is why a fall in price is a gain for buyers as a group.

Limits of the idea.

The measure has limits. It assumes that money has the same marginal utility for everybody, which is not exactly true. It cannot be applied to necessities, where the first unit is worth more than any sum. And it assumes there is no close substitute. Even so, it is a useful guide, and the next paragraph shows where.

Producer's surplus the difference between the price a seller receives and the lowest price the seller would accept.

The sellers' gain, as the mirror image.

Producer's surplus is the sellers' counterpart: the difference between the price received and the lowest price the seller would have accepted. Together, consumer's and producer's surplus measure the gain from trade. Governments use them to compare policies: a tax, a price ceiling or a monopoly reduces the sum, and the lost part is called the welfare loss. Public bodies use consumer's surplus to price water and power, and to weigh a new bridge against its cost.

CONSUMPTION AND CHOICE

Consumption the use of goods and services to satisfy wants.

Consumption ends the chain of production.

Consumption means the use of goods and services to satisfy wants. A consumer's budget is limited, so the consumer must choose among goods. Utility counting, with its utils, is hard to defend. The next method needs only ranking.

INDIFFERENCE CURVE ANALYSIS

Indifference curve a line showing all combinations of two goods that give a consumer equal satisfaction.

Every point on one curve is equally good.

Four properties.

An **indifference curve** means a line that joins all combinations of two goods that give the consumer the same satisfaction. If you are equally happy with 4 buns and 2 cups of tea, or 2 buns and 5 cups of tea, both points lie on one curve. The consumer is "indifferent" between them. Edgeworth (1881), Pareto (1906) and Hicks and Allen (1934) developed the tool.

Indifference curves have four properties. They slope downward, because more of one good must replace less of the other. They are convex to the origin, because of a falling **marginal rate of substitution** (MRS), the amount of one good given up for one more of the other. They never cross, since crossing would make one bundle both better and equal. And a curve higher and further from the origin shows greater satisfaction.

A consumer cannot choose any point she likes. The **budget line** shows all combinations she can buy with her income at given prices. Suppose income is Tk 120, a bun costs Tk 10 and a cup of tea costs Tk 20. She can buy 12 buns and no

Budget line all combinations of two goods that the consumer's income can buy at given prices.

Income and prices draw the budget line.

THINK IT THROUGH. Supply rises by 30 at every price, so $Q_s = 6P - 30$. Using $Q_d = 300 - 6P$, find the new price and quantity.

WHERE PEOPLE GO WRONG. "Equilibrium means everyone is happy." It means only that plans match. Some buyers still cannot afford the good.

IN HISTORY. In the 1890 edition of his book, Marshall used diagrams where demand and supply cross, and called the crossing point the equilibrium. Before him, writers argued which one set the price.

REVIEW QUESTIONS

1. Define equilibrium price.
2. At a price above equilibrium, what exists and what happens to price?
3. $Q_d = 100 - 2P$, $Q_s = 3P - 20$. Find P .



CHAPTER THE EIGHTH

PRODUCTION AND RETURNS

What happens when we add one more baker?

The chapter begins with a bakery that hires workers one by one. It defines the production function, shows why extra workers add less and less, and distinguishes this from returns to scale.

The bakery has one oven. With one baker it makes 40 loaves a day. A second baker makes it possible to bake 100. A third raises output to 180. A seventh baker, however, makes output fall. Why should a helper ever reduce output? The answer explains the shape of every cost curve in the next chapter.

Start with a question you can almost answer.

THE PRODUCTION FUNCTION

Production means the making of goods and services that satisfy wants. Economists group the resources used into four **factors of production**: land (all natural resources), labour (human effort), capital (machines, buildings and tools) and enterprise (the skill of organising and taking risk).

The **production function** means the relationship between the inputs used and the greatest output they can make. In words: output depends on land, labour, capital and enterprise. In symbols, $Q = f(L, K)$, where Q is output, L is labour and K is capital. In the **short run**, at least one input is fixed. In the **long run**, all inputs can change.

Production the creation of goods and services that satisfy wants.

Production creates utility.

Production function the relation between the inputs used and the maximum output obtained.

Inputs in, output out.

THE LAW OF DIMINISHING RETURNS

Take the short run, with the oven fixed and labour variable. **Total product** (TP) is the total output. **Average product** (AP) is output per worker, TP divided by workers. **Marginal product** (MP) is the extra output from one more worker.

Three measures.

OUTPUT OF THE BAKERY WITH ONE OVEN

Bakers	Total product	Average product	Marginal product
1	40	40	40
2	100	50	60
3	180	60	80
4	240	60	60
5	280	56	40
6	300	50	20
7	290	41	-10

quantity does not change at all) or **perfectly elastic** (Ed is infinite, buyers will buy any amount at one price and none above it).

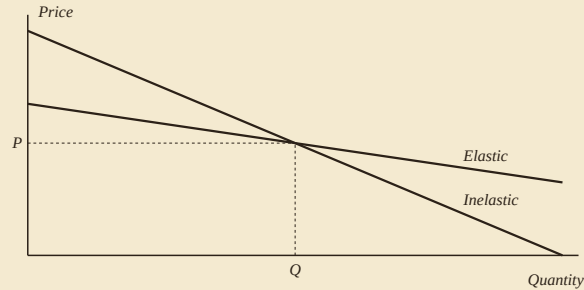


FIG. 4. Two demand curves through the same point. The flatter one is more elastic.

What decides elasticity?

Four factors matter. The more substitutes a good has, the more elastic its demand: if tea is dear, buyers switch to coffee. Necessities, such as salt, have inelastic demand; luxuries have elastic demand. A good that takes a large share of income is more elastic than one that takes a small share. And demand is more elastic over a long period, because buyers have time to find other goods.

INCOME AND CROSS ELASTICITY

Income elasticity of demand the percentage change in quantity demanded divided by the percentage change in income.

Income elasticity links buying to income.

Cross elasticity of demand the percentage change in quantity of one good divided by the percentage change in the price of another good.

Cross elasticity shows how goods are related.

Income elasticity of demand means the percentage change in quantity demanded divided by the percentage change in income. If income rises 10 per cent and cake purchases rise 20 per cent, the value is 2. A value above 1 marks a luxury; between 0 and 1 marks a necessity; below 0 marks an **inferior good**, one that people buy less of as they grow richer, such as very cheap coarse rice.

Cross elasticity of demand means the percentage change in the quantity of one good divided by the percentage change in the price of another. If a rival's bread price rises 10 per cent and our sales rise 5 per cent, the value is 0.5. A positive value shows that the goods are substitutes. A negative value shows that they are complements. A value near zero shows that the goods are unrelated.

WHY ELASTICITY MATTERS

Total revenue means price multiplied by quantity sold. With elastic demand, a price cut raises revenue, because quantity rises by more than price falls. With inelastic demand, a price cut lowers revenue. A firm that knows its elasticity can choose the right price. So a seller of an inelastic good, such as salt, can raise its price with little loss of sales; a seller of an elastic good, such as cake, cannot.

CASE FILE: PRICE CHANGE TEST, 1 APRIL 2024

The bakery moves the loaf price from Tk 30 to Tk 35. Revenue before: $30 \times 120 =$ Tk 3,600. Revenue after: $35 \times 90 =$ Tk 3,150. Revenue falls, as the elasticity of 1.5 predicts. The bakery returns to Tk 30.

SUMMARY. Elasticity measures how much quantity responds to a change in price (or income, or another price). Demand is elastic above 1, inelastic below 1. Substitutes, necessity, share of income and time decide its size. With elastic demand, a price cut raises revenue.

CASE FILE: STAFFING, 1 SEPT 2024

The bakery sets four bakers, since MP is still 60 and AP is at its highest (60). A fifth baker would add only 40 loaves, which is less than the average. The owner leaves the number at four.

SUMMARY. The production function links inputs to output. In the short run, extra units of a variable input eventually add less. In the long run, doubling all inputs may more than double, exactly double or less than double output.

THINK IT THROUGH. Using the table, find MP of the fifth baker, and AP with six bakers. Why does the owner stop at four, and not seven?

WHERE PEOPLE GO WRONG. "Diminishing returns means total output falls." Not at first. Total product still rises while MP is positive.

IN HISTORY. In 1767 the French minister Turgot saw that adding more labour to a fixed plot of land gave smaller and smaller extra harvests. Ricardo used the same idea to explain rent.

REVIEW QUESTIONS

1. Define the production function.
2. Distinguish diminishing returns from returns to scale.
3. Inputs double and output rises 60%. What returns?



CHAPTER THE NINTH

THE COSTS OF PRODUCTION

What does a loaf really cost?

The chapter begins with a baker who thinks her own time is free. It separates explicit and implicit costs, builds the cost curves from a table, compares short and long run, and finds the cheapest mix of inputs.

The owner of Rupali Bakery adds up her bills for flour, rent, wages and gas, and calls the total her cost. She has left out her own work and the money she put into the shop. Is her real cost larger than the bills? Economists say yes, and the difference changes the answer to the question “Am I making a profit?”

Start with a question you can almost answer.

OPPORTUNITY COST AND KINDS OF COST

Recall from Chapter 1 that **opportunity cost** is the value of the best alternative given up. To measure it, ask what the same resources would earn in their next best use. In the bakery, one cake needs the oven for as long as five loaves. So the opportunity cost of one cake is 5 loaves, and of one loaf it is 1/5 of a cake, that is, 0.2 cake. The cost is stated in goods, not only in money.

Opportunity cost, measured.

An **explicit cost** is a payment to an outsider, such as for flour or rent. An **implicit cost** is the value of the owner’s own resources that are used in the business. If the owner could earn Tk 1,000 a day in another job, that sum is an implicit cost of her time. If her savings could earn Tk 500 a day elsewhere, that is an implicit cost of her capital. Two examples of implicit cost are therefore the owner’s time and the owner’s own money.

Explicit cost a cost that needs a payment to someone outside the firm.

Implicit cost the value of resources the owner supplies, for which no payment is made.

Two kinds of cost: paid and unpaid.

Costs also differ by how they react to output. A **fixed cost** (FC) does not change with output in the short run, such as rent and the oven. A **variable cost** (VC) changes with output, such as flour and gas. **Total cost** (TC) is fixed plus variable cost. Each cost can also be stated per unit: **average cost** (AC) is TC divided by output, and **marginal cost** (MC) is the extra cost of one more unit.

Fixed cost a cost that does not change with output in the short run.

Variable cost a cost that changes as output changes.

Fixed does not move; variable does.

COSTS OF THE BAKERY, IN TAKA PER DAY (FC = 1,500)

Loaves	VC	TC	AC	AVC	MC
100	2,400	3,900	39.0	24.0	24
200	4,200	5,700	28.5	21.0	18
300	5,700	7,200	24.0	19.0	15
400	7,600	9,100	22.8	19.0	19
500	10,500	12,000	24.0	21.0	29

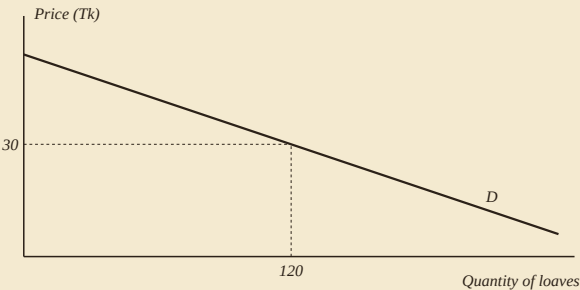


FIG. 3. The demand curve for loaves. It slopes downward from left to right.

Reading the figure.

To read the curve, pick a price on the vertical axis and go across to the line, then down to the quantity. At Tk 30 the quantity is 120 loaves. Moving along the curve, because of a change in price, is a **change in quantity demanded**. This is different from a shift of the whole curve.

FROM ONE BUYER TO THE MARKET

Adding up buyers.

A **market demand** curve is the sum of the demands of all buyers at each price. If two customers each buy 60 loaves at Tk 30, market demand at Tk 30 is 120 loaves. A price rise therefore reduces quantity both because each buyer buys less and because some buyers drop out. Throughout this book, “demand” means market demand unless we say otherwise.

Substitute goods goods that can replace each other, such as tea and coffee.

Complementary goods goods used together, such as bread and butter.

Related goods change demand.

Goods can be related. **Substitute goods** can replace each other, so a rise in the price of one raises demand for the other. **Complementary goods** are used together, so a rise in the price of one lowers demand for the other. Suppose the price of petrol rises. People drive less, and demand for cars, which are used with petrol, probably falls. Demand for bicycles, a substitute, probably rises.

SHIFTS OF THE CURVE

Other things that move demand.

The demand curve is drawn for given values of other things: the buyer’s income, tastes, the prices of related goods, expectations and the number of buyers. When one of these changes, the whole curve shifts. This is a **change in demand**. A rise in income usually moves the curve to the right: buyers want more at every price. A fall moves it to the left. Remember the difference: price moves us along the curve; every other cause shifts it.

EXCEPTIONS TO THE LAW

Giffen good a very basic good for which demand rises when its price rises.

Rare, but a real exception.

Other exceptions.

The law has a few exceptions. A **Giffen good** is a basic food for which quantity demanded rises when the price rises. Marshall (1895) credited Robert Giffen with the idea and linked it to the potato in nineteenth-century Ireland, though modern historians dispute the evidence. The reasoning is this: for a very poor family, a basic food takes most of its income. When its price rises, the family cannot afford better food, and it buys more of the basic food to stay fed.

Two other cases concern pride and crowds. The **snob effect** is the wish to own a good because few others have it: a rise in price may make a rare watch more desirable. The **bandwagon effect** is the wish to own what others own:

WORKED EXAMPLE. Least-cost mix

GIVEN	A baker’s wage is Tk 800 a day; a machine hire is Tk 400 a day. Extra output: baker 40 loaves; machine 24 loaves.
FIND	Which input is the better buy.
FORMULA	MP of labour / wage compared with MP of capital / hire price.
A	
STEPS	Baker: 40 / 800 = 0.05 loaf per taka. Machine: 24 / 400 = 0.06 loaf per taka.
ANSWER	Machines give more per taka, so hire more machines and fewer bakers.
CHECK	Check: as machines increase, their MP falls (diminishing returns), until the two ratios are equal. Then no further change pays.

CASE FILE: COSTS, 1 OCT 2024

At 500 loaves a day: total cost Tk 12,000 (explicit). Implicit costs: owner’s time Tk 1,000, own capital Tk 500. Full economic cost: Tk 13,500.

SUMMARY. Opportunity cost is the best alternative given up. Costs may be explicit or implicit, fixed or variable. AC and MC are U-shaped; MC cuts AC at its minimum. The long-run curve allows all inputs to change. The cheapest mix equalises extra output per taka.

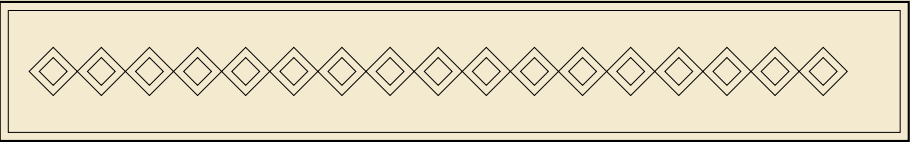
THINK IT THROUGH. If FC is Tk 1,500 and VC at 200 loaves is Tk 4,200, find AFC, AVC and AC. Which cost is the owner’s own time?

WHERE PEOPLE GO WRONG. “Sunk and fixed costs should change the price.” In the short run, fixed costs do not change with output, so they do not affect the choice of how much to bake.

IN HISTORY. The term “opportunity cost” was first used by Friedrich von Wieser in 1914. Before him, Ricardo had already compared the cost of a good with the other goods that the same labour could make.

REVIEW QUESTIONS

1. What is opportunity cost?
2. Name two implicit costs.
3. AC = 24, MC = 29. Does AC rise or fall?



CHAPTER THE TENTH

MARKET STRUCTURES AND THE FIRM

Who decides the price: the seller or the market?

The chapter begins with a baker who must accept the price and a monopolist who may choose it. It defines market structure, derives equilibrium for the competitive firm and industry, and compares monopoly and monopolistic competition.

The bakery cannot sell a loaf at Tk 50 when the shop next door sells at Tk 30, because buyers walk away. The only water company in a small town can raise its price and still sell. What separates the two sellers? Not their wish for profit, but the number of rivals they face and how alike their goods are.

Start with a question you can almost answer.

MARKET STRUCTURE

Market structure means the features of a market that decide how firms behave: the number of sellers, whether the goods are alike, and how easily new firms can enter. Four main types exist. Price-setting power rises as we move down the table.

Market structure the features of a market that decide how firms behave, mainly the number of sellers, the kind of product and the ease of entry.
Competition decides who sets the price.

FOUR MARKET STRUCTURES

Type	Sellers	Product	Entry	Control of price
Perfect competition	Very many	Identical	Free	None
Monopolistic competition	Many	Differentiated	Free	Some
Oligopoly	Few	Alike or differentiated	Hard	Considerable
Monopoly	One	No close substitute	Blocked	Large

PERFECT COMPETITION

Perfect competition means a market with very many small buyers and sellers, identical goods, free entry and exit, and full knowledge of prices. **Pure competition** is a narrower idea: it needs only the first three, with no element of monopoly, and it does not need full knowledge or free movement of resources. In perfect competition, each firm is a **price taker**. It accepts the market price, so its extra revenue from one more unit (marginal revenue, MR) equals the price.

Perfect competition a market with very many small buyers and sellers, identical goods, free entry and full information.
Price is set by the market.

said a person can only rank satisfactions as first, second and third. This is the **ordinal** view, and we use it in Chapter 5.

THE LAW OF DIMINISHING MARGINAL UTILITY

Law of diminishing marginal utility as a person consumes more units, the extra satisfaction from each further unit falls.

The more we have, the less the next one adds.

Look at the example again. The marginal utility falls with each bun: 20, 16, 12, 8, 4, 0. This is the **law of diminishing marginal utility**. It states that, as a person consumes more units of a good in a given period, the marginal utility of each extra unit falls. Total utility still rises while marginal utility is positive. It reaches its highest point when marginal utility is zero, here after the fifth bun.

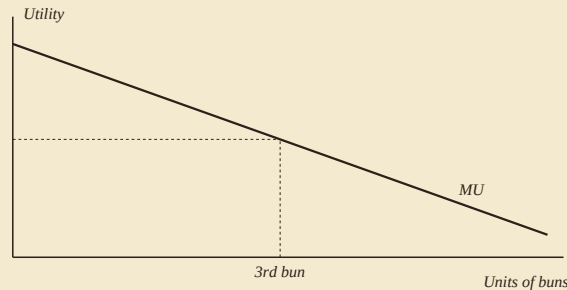


FIG. 2. Marginal utility falls as units rise (the schedule of buns, drawn smoothly).

Reading the figure.

To read the figure, find a number of buns on the horizontal axis and go up to the line. The height shows the extra satisfaction from that bun. The line slopes down, which is the law in picture form. Where the line meets the horizontal axis, the extra satisfaction is zero, and the student should stop eating.

The assumptions behind the law.

The law holds under conditions. The units must be of the same kind and of reasonable size. Consumption must be continuous, with no long breaks. The consumer's taste, income and the prices of related goods must stay the same. The consumer must act sensibly. If a condition fails, the law may appear to fail too.

Limits and exceptions.

Some cases seem to break the law. The urge for a hobby, such as collecting stamps, may grow with each stamp. A person who loves money may want more money as the pile grows. A drug user may want ever more of the drug. Knowledge may bring more interest in further knowledge. Most economists answer that in these cases a condition fails: the units are not of the same kind, the taste is changing, or the consumption is not rational. Even then, at some point the satisfaction from the next unit falls.

Back to the puzzle.

Now return to water and diamonds. Water is so plentiful that its marginal utility is low. Diamonds are rare, so the marginal utility of one more diamond is high. Price follows marginal utility, not total utility. Water has a very high total utility, but its price is low. In 1871 Jevons and Menger, and in 1874 Walras, each gave this answer separately.

CASE FILE: CUSTOMER SURVEY, 20 JAN 2024

Rupali Bakery asks one customer how much she would pay for each bun in a day. Answers: Tk 20 for the first, 16 for the second, 12 for the third, 8 for the fourth, 4 for the fifth. These are her marginal utilities in money. At a price of Tk 10 she buys three buns.

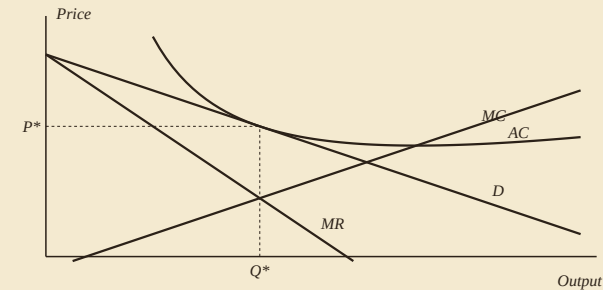


FIG. 11. Long-run equilibrium under monopolistic competition. The demand curve (D) touches AC at one point, and MR meets MC at the same output.

MARKET STRUCTURE AND PRICING

Structure decides whether a firm takes the price or makes it. A perfect competitor takes it. A monopolist and a monopolistic competitor choose it, within the limit of demand. An **oligopolist**, one of a few sellers, must also guess how rivals will answer a price change, so prices tend to stay stable. Many firms aim for goals other than profit, such as sales, market share or growth. Profit is still the most useful single guide, but it is not the only goal.

The role of structure in pricing.

CASE FILE: MARKET POSITION, 1 JAN 2025

The bakery sells a loaf with its own recipe, and the shop two streets away sells a similar one. Rupali Bakery therefore faces monopolistic competition. It has some control over price, and any profit above normal draws in new bakeries.

SUMMARY. Market structure depends on number of sellers, kind of product and entry. A competitive firm sets $P = MC$; entry drives profit to normal in the long run. A monopolist sets $MR = MC$ and charges a price above MC . Monopolistic competition has differentiation and, in the long run, normal profit and excess capacity.

THINK IT THROUGH. At Tk 30, would the firm of Chapter 9 bake 300 loaves? Compare price with MC at 300 and say whether more or less output would raise profit.

WHERE PEOPLE GO WRONG. "Perfect and pure competition are the same." They are not. Perfect competition adds full knowledge and perfect mobility to the conditions of pure competition.

IN HISTORY. In 1933, two books appeared in the same year, by Edward Chamberlin in the United States and Joan Robinson in England. Each showed that most real firms are neither perfect competitors nor pure monopolists.

REVIEW QUESTIONS

1. Define market structure.
2. State the profit-maximising rule of a competitive firm.
3. What is product differentiation?

Is economics a science or an art?

A **science** is an organised body of knowledge that links causes to effects. An **art** is the skilful use of knowledge to reach a goal. Economics is both. It is a science because it builds laws from observation and reasons from cause to effect. It is an art because a baker, a banker or a minister applies those laws to reach a goal. Positive economics is the science, and normative economics guides the art.

CASE FILE: DECISION ON 1 JANUARY 2024

The owner chooses 300 loaves and 16 cakes for the first day. Check against the oven: 300 loaves use 6 of 8 oven hours; 16 cakes use 1.6 hours; total 7.6 hours. The remaining 0.4 hour is unused, which is waste in the language of this chapter.

SUMMARY. Scarcity forces choice. Economics studies how people choose when resources are limited. Micro studies single decision-makers and markets; macro studies the whole economy. Economic laws state tendencies that hold when other things are equal. Positive statements describe; normative statements judge.

THINK IT THROUGH. A student can spend Saturday working for Tk 600 or studying for a test. She chooses to study. State what she gave up and why this is a choice caused by scarcity.

WHERE PEOPLE GO WRONG. “Scarcity means poverty.” It does not. Even the richest country cannot satisfy every want. Also, a normative opinion is not wrong because it is an opinion; it is simply not a positive statement.

IN HISTORY. In 1776 Adam Smith described a pin factory in which ten workers, each doing one task, made about 48,000 pins a day. Working alone, each could hardly make twenty. He used the case to show that dividing tasks raises output from the same scarce labour.

REVIEW QUESTIONS

- 1. Why does scarcity not mean poverty?
- 2. Give one micro and one macro question.
- 3. Classify: “Bread prices should be capped.” Positive or normative? Why?



CHAPTER THE ELEVENTH

RENT, WAGES, INTEREST AND PROFIT

Who gets what the bakery earns?

The chapter begins with the bakery’s daily takings and asks who receives them. It explains how rent, wages, interest and profit are fixed, using Ricardo’s theory, the marginal productivity idea and the main theories of profit.

On a good day Rupali Bakery takes Tk 15,000. Part goes to the landlord, part to the bakers, part to the lender who financed the oven, and part is left for the owner. What decides each share? Economists call this question **distribution**: the sharing of the product among the owners of land, labour, capital and enterprise.

Start with a question you can almost answer.

RENT

Economic rent means the payment to a factor whose supply is fixed, in excess of what is needed to keep it in use. Land is the main example, since the nation’s land cannot be increased by a higher price. Ricardo (1817) built a theory on it. Rent is different from the everyday word, which means a payment to a landlord for a house or shop.

Economic rent the payment to a factor in fixed supply, above what is needed to keep it in use. Rent is the reward for what is in fixed supply.

WORKED EXAMPLE. Ricardo’s theory

GIVEN	Three plots of equal size, each worked with the same cost. Wheat yields: plot A 40, plot B 30, plot C 20 units.
FIND	The rent of each plot.
FORMUL A	Rent = yield of the plot minus yield of the marginal (worst) plot used.
STEPS	Plot C is the marginal plot: it only covers its costs, so its rent is 0. Rent of B = 30 - 20 = 10. Rent of A = 40 - 20 = 20.
ANSWER	Rents: A 20, B 10, C 0 units of wheat.
CHECK	Check: rent measures the extra advantage over the worst land. A has the largest advantage (20), so it has the largest rent.

Ricardo said that rent is a **differential surplus**: it arises from differences in fertility and location. On the marginal land there is no rent, so rent does not enter the price of grain. Critics reply that land has alternative uses, so for the economy as a whole it does have a cost. The modern view agrees with Ricardo for a single use, and adds that rent can be viewed as a cost for a user who must outbid others. **Quasi-rent** is a temporary surplus on a man-made asset, such as a machine, until supply can be increased. **Artificial rent** is a surplus created by restrictions, such as a licence, and it differs from the real rent of land.

Differential surplus the extra produce of better land over the worst land in use. Rent is a surplus, not a cost on the margin.

Opportunity cost the value of the best alternative given up when a choice is made.

The real cost of a choice is what is given up.

ignored well-being, and that Marshall's definition left out services that do not add to welfare. Robbins's definition covers both, but it says little about growth over time. For a first course, remember this: economics is about choice made necessary by scarcity.

The baker's problem shows a key idea. If she bakes 80 cakes, she gives up 400 loaves. The loaves given up are the **opportunity cost** of the cakes. Opportunity cost means the value of the best alternative that you give up when you choose. We study it fully in Chapter 9. For now, hold the idea: choosing one thing means not choosing another.

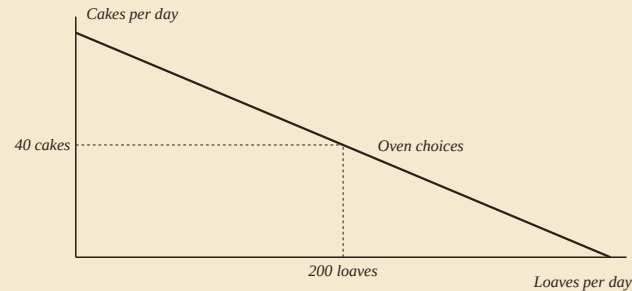


FIG. 1. Every point on the line is one possible day's baking. The point shown is a mixture.

Reading the figure.

Read the figure in three steps. The horizontal axis counts loaves, and the vertical axis counts cakes. The line joins the two extreme choices: all cakes at the top, all loaves at the right. A point on the line, such as 200 loaves and 40 cakes, shows a mixture. To gain more cakes, you must move left, and that means fewer loaves.

You should now be able to say: "Resources are scarce, so a choice for one thing is a choice against another." Test yourself before you read on.

THE BASIC PROBLEMS OF AN ECONOMY

Every society, rich or poor, must answer three questions. **What** goods shall be produced, and in what amounts? **How** shall they be produced, by hand or by machine, in small shops or large factories? **For whom** shall they be produced, that is, how shall the output be shared? The baker answers all three for her shop. A country answers them for millions of people, through its **economic system**.

Two more problems follow. Society must use its resources fully and without waste, and it must decide how much to save for the future so that output can grow. Unused ovens and idle workers are a waste. Both problems still rest on scarcity. Markets, governments or a mixture of both give the answers. Bangladesh is a mixed economy: private firms and the state both take part.

MICRO AND MACRO

Microeconomics the study of the choices of single households, firms and markets.

Micro looks at the parts.

Macroeconomics the study of the whole economy, such as total output, prices and jobs.

Macro looks at the whole.

Four main theories answer. Knight (1921) said profit rewards the bearing of **risk**, the uncertainty that cannot be insured. The marginal productivity theory treats the entrepreneur as one more factor, paid for the extra output she brings. Schumpeter (1911) said profit rewards **innovation**, the new method or product, and falls away as rivals copy it. A fourth view is that monopoly power or luck may also give surplus profit, which does not reward effort.

Why profit occurs.

CASE FILE: THE DAY'S TAKINGS, 1 OCT 2024

Takings Tk 15,000; materials, gas and other running costs; four bakers at Tk 800 = 3,200; rent; interest on the oven loan; and the rest to the owner. The owner's economic profit is Tk 1,500, as computed above.

SUMMARY. Rent rewards a factor in fixed supply and is a differential surplus on land. Wages depend on marginal productivity, supply and bargaining; real wages correct for prices. Interest rewards waiting and is set by loanable funds and liquidity. Profit rewards risk, innovation and enterprise, and economic profit is smaller than accounting profit.

THINK IT THROUGH. Plot A yields 50, B 35 and C 25, all at the same cost. Find the rent of A and B by Ricardo's method.

WHERE PEOPLE GO WRONG. "A higher nominal wage means a higher real wage." Not if prices rise faster. Always compare the two percentage changes.

IN HISTORY. In 1815 Ricardo wrote about the Corn Laws, the duties on imported grain in Britain, and argued that they raised rents of landowners at the cost of the rest. He used the theory of rent to make the case.

REVIEW QUESTIONS

1. Define economic rent.
2. Distinguish nominal wage from real wage.
3. Revenue 10,000, explicit costs 7,000, implicit costs 1,200. Find both profits.

TO THE READER

This book teaches **microeconomics**, the study of how people, households and firms make choices and how those choices meet in markets. You need no earlier knowledge of business. You need only arithmetic and basic algebra.

How a page is built. The wide column holds the main text. The narrow column at the outer edge holds notes. An *italic* note gives the one-line point of the paragraph beside it. A note that begins with a word in **bold type** gives the plain meaning of a key term.

Bold type in the main text marks a key term at its first use only. Italics are used for emphasis, for titles of books and for foreign words. Each key term is explained in simple English, then shown in an example. A full list appears at the back, in the Glossary and Index, with the page of first use.

How a chapter is built. Each chapter opens with a question. Each idea then follows four steps: the idea, an example, the rule, and the exceptions. A **worked example** shows every step of a calculation in the same order: Given, Find, Formula, Steps, Answer, Check. A **case file** follows one small bakery through the book. The bakery is invented for teaching.

How to study. Read slowly. Where a passage ends with “You should now be able to say”, test yourself aloud before you move on. Do the review questions at the end of each chapter, then check your answers at the back.

The mathematics. Percentage change means the change divided by the starting value, multiplied by 100. A ratio compares two numbers by division. To read a graph, find a point on the horizontal axis, go up to the curve, then read across to the vertical axis. Nothing else is needed.

Market structure: the features of a market that decide how firms behave, mainly the number of sellers, the kind of product and the ease of entry. *Ch. 10, p. 41.*

Microeconomics: the study of the choices of single households, firms and markets. *Ch. 1, p. v.*

Monopolistic competition: a market with many sellers of similar but not identical goods, and free entry. *Ch. 10, p. 42.*

Monopoly: a market with a single seller of a good that has no close substitute. *Ch. 10, p. 42.*

Nominal wage: the wage paid in money. *Ch. 11, p. 46.*

Opportunity cost: the value of the best alternative given up when a choice is made. *Ch. 1, p. v.*

Perfect competition: a market with very many small buyers and sellers, identical goods, free entry and full information. *Ch. 10, p. 41.*

Positive statement: a statement about what is, which facts can test. *Ch. 1, p. vi.*

Price elasticity of demand: the percentage change in quantity demanded divided by the percentage change in price. *Ch. 4, p. 17.*

Price elasticity of supply: the percentage change in quantity supplied divided by the percentage change in price. *Ch. 6, p. 26.*

Producer’s surplus: the difference between the price a seller receives and the lowest price the seller would accept. *Ch. 5, p. 22.*

Production: the creation of goods and services that satisfy wants. *Ch. 8, p. 33.*

Production function: the relation between the inputs used and the maximum output obtained. *Ch. 8, p. 33.*

Real wage: what the money wage can buy. *Ch. 11, p. 46.*

Returns to scale: the change in output when all inputs are changed in the same proportion. *Ch. 8, p. 34.*

Scarcity: resources are limited while wants are not. *Ch. 1, p. iv.*

Substitute goods: goods that can replace each other, such as tea and coffee. *Ch. 3, p. 14.*

Supply: the quantity of a good that sellers are willing and able to offer at each price in a given period. *Ch. 6, p. 25.*

Total utility: the whole satisfaction from all the units consumed. *Ch. 2, p. viii.*

Utility: the satisfaction a person gets from using a good. *Ch. 2, p. viii.*

Variable cost: a cost that changes as output changes. *Ch. 9, p. 37.*

FURTHER READING

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MICROECONOMY

Prices, Markets, Costs and Rewards, explained from first principles

A Primer for First-Year Students of Business Studies

B I J O Y ? !

British spelling. Currency: the Bangladeshi taka (Tk). The bakery used in the examples is *illustrative*: it is not a real business. All figures in the examples are invented to teach method.